

Name: _____ score: _____ / 10

There are 10 points possible on this quiz. No aids (book, notes, etc.) are permitted. You may use a non-programmable calculator. **Show all work and supporting calculations for full credit. Explain how you get your answers.**

1. (3 points) Consider the weighted voting system $[31 : 16, 12, 10, 3]$

(a) What is the meaning of the number **31** in the weighted voting system?

It is the quota = # votes needed to take action

(b) Why is $\{P_1, P_2, P_4\}$ a **winning coalition**? Explain your answer with a calculation and a few words.

total weight = $16 + 12 + 3 = 31 \geq 31$
so together they can achieve quota

(c) Which players, if any, have **veto power**? Explain your reasoning.

$12 + 10 + 3 = 25 < 31$ so P_2, P_3, P_4 can't make quota
So $\boxed{P_1}$ is a member of every winning coalition and
therefore has veto power.

2. (2 points) Consider the weighted voting system $[q : 12, 8, 2, 1]$.

(a) What is the **smallest possible quota value**? Explain how you know.

$12 + 8 + 2 + 1 = 23$ and $23/2 = 11.5$

So since you must have a majority to be an allowable quota
we need $\boxed{q > 12}$

(b) Determine a value of q that results in a **dictator**. Explain which player is the dictator and how you know.

$q = 12$ works, since it's an allowable quota value
and if $q = 12$ then P_1 can make quota alone.

3. Consider the weighted voting system $[17 : 13, 9, 6, 2]$ and the list of all possible coalitions.

coalition	weight	winning?	coalition	weight	winning?
<u>P_1</u> <u>P_2</u>	22	✓	<u>P_1</u> P_2 P_3	28	✓
<u>P_1</u> <u>P_3</u>	19	✓	<u>P_1</u> <u>P_2</u> P_4	24	✓
P_1 P_4	15	X	<u>P_1</u> <u>P_3</u> P_4	21	✓
P_2 P_3	15	X	<u>P_2</u> <u>P_3</u> <u>P_4</u>	17	✓
P_2 P_4	11	X	P_1 P_2 P_3 P_4	30	✓
P_3 P_4	8	X			

- (a) (1 point) In the table above, write the weight of each coalition in the appropriate column.
- (b) (1 point) Put a check ✓ in the column labeled “winning?” to identify all winning coalitions.
- (c) (2 points) In each winning coalition, underline the critical players.
- (d) (1 point) Are there any dummies? Explain why or why not, with a sentence or short computation and sentence.

No — every player (even P_4) shows up in at least one winning coalition.